

Facade Material Detection from Building Photos

M4U3 Computer Vision · Mini report · Omar Elsayed · 27 September 2026 ·
github.com/OmarEAbdelaal/M4U3-CV-Facade-Materials

1. Executive summary

Facade materials fixed by value-engineering decisions during design are rarely verified once the building is under construction. This prototype detects six facade materials (brick, cladding panel, exposed concrete, glass, painted render, stone cladding) in building photos, so that a mismatch with the specification can be flagged for a person to review. A YOLOv8s model, fine-tuned from COCO on 229 openly licensed Wikimedia Commons photos, reaches **validation mAP@50 0.394, precision 0.708 and recall 0.345** (test: 0.315, 0.480, 0.319). Precision meets the 0.50 prototype target; recall and mAP@50 do not. The model is reliable when it fires but misses 42–77% of each material, so it is a proof of concept, not yet a usable screen. The largest improvement came from the data, not the model: cleaning the labels raised mAP@50 from 0.14 to 0.39 with identical training settings.

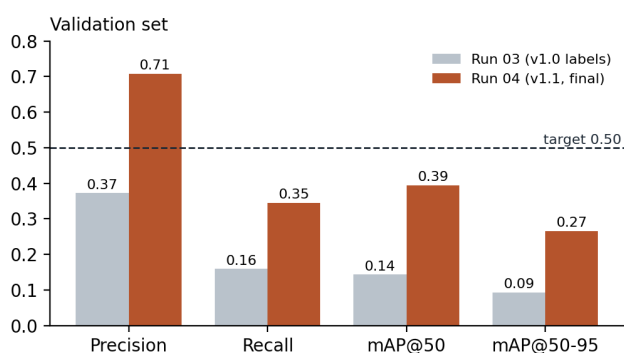
2. Data and method

Images were collected from Wikimedia Commons with a license filter (CC0, public domain, CC BY, CC BY-SA) and labelled in Roboflow with boxes and SAM polygons. Inspection of the export found four label problems: polygons mixed with boxes in 181 files, about 3,500 fragment boxes, glass and stone labelled per pane or per stone instead of per material area, and nested or stacked labels. A deterministic cleaning notebook converts polygons to boxes, removes fragments, merges each continuous same-class area into one box, resolves stacked classes and removes boxes under 1% of the image, giving Release v1.1 (322 / 45 / 23 images, 968 / 149 / 78 boxes). The dataset is frozen as a GitHub Release with a SHA256 checksum, so the notebook runs in Colab with no accounts or keys. Training: YOLOv8s, 50 epochs, batch 16, 640 px, seed 0, ultralytics 8.4.163, Tesla T4, 5.7 minutes.

3. Results

Split / class	Precision	Recall	mAP@50	mAP@50-95
Validation, all	0.708	0.345	0.394	0.265
Test, all	0.480	0.319	0.315	0.248
val · brick	0.776	0.318	0.383	0.180
val · cladding_panel	0.474	0.231	0.276	0.221
val · exposed_concrete	0.853	0.530	0.628	0.485
val · glass	0.472	0.265	0.294	0.203
val · painted_render	0.895	0.250	0.310	0.238
val · stone_cladding	0.778	0.474	0.472	0.262
Run 03 (uncleaned labels), validation	0.372	0.159	0.143	0.093

The normalised confusion matrix shows the share of each class predicted as background (missed): brick 0.55, cladding_panel 0.77, exposed_concrete 0.45, glass 0.63, painted_render 0.56, stone_cladding 0.42. Of the detections on unlabelled areas, 66% are glass. Painted render is confused with brick in 12% of cases. Run 03 and Run 04 are scored on different answer keys (v1.0 vs v1.1), so part of the gain comes from a more consistent key.



4. Error analysis

- **FP1** — glass fired on unlabelled areas (66% of background detections). Cause: mostly real windows missing from partly labelled images (label gap A6).
- **FP2** — stone_cladding on roof edges and sills (11% of background detections). Cause: some stone labels outline roofs (A4).
- **FP3** — painted_render called brick in 12% of cases. Cause: warm-coloured render with joints; few render examples after cleaning.
- **FN1** — cladding_panel: 77% missed. Cause: fewest examples (13 validation boxes) and very varied metal finishes that look like glass or sky.
- **FN2** — glass: 63% missed. Cause: two object scales after merging (curtain wall vs punched windows) and sky reflections.

- **FN3** — brick on the test split: recall 0.06. Cause: frame-filling walls under-represented (E1); only 7 test images.

Next data improvements, by impact: (1) relabel to one standard, starting with the 24 images removed in cleaning and the partly labelled ones, and review stone_cladding roofs; (2) at least 80 images per class, most for cladding_panel and painted_render, including frame-filling walls; (3) hard cases on purpose (glare, shadow, occlusion, look-alike materials), with black-and-white images kept out of training.

5. Limitations and governance

- **Screening aid, not certification.** Output is a flag for a qualified person. It must not certify compliance, approve payment or handover, or judge fire and life safety: appearance says nothing about core material, rating or fixings.
- **False negatives cost more than false positives:** a missed material is a substitution nobody reviews. Current recall (0.35) is the main limitation; in use, a lower confidence threshold should trade precision for recall.
- **Data limits:** 229 images, small validation and test splits (13–68 boxes per class), labels enforced in code but not fully relabelled; performance on night, glare, B&W or distant photos is unknown. Look-alike materials (stone-look panels, brick slips) are out of scope.
- **Privacy and licences:** only public, openly licensed photos; no client or site images; faces and plates to be blurred in the next version. Code MIT; dataset CC BY-SA 4.0 with per-image attribution; weights AGPL-3.0 (Ultralytics).

6. Next steps

Relabel and publish v1.2; retrain and choose the threshold on validation for recall ≥ 0.75 on glass, cladding_panel, stone_cladding and exposed_concrete; then connect detections to IFC material parameters so a site photo can be checked against the value-engineering decision log in the Final Master's Project.